ClimateScope:Visualizing Global Weather Trends and Extreme Events

Milestone 3: Visualization Development & Interactivity Report

1. Introduction

This report summarizes the development of the interactive dashboard for the ClimateScope project as part of Milestone 3. The primary objective of this phase was to transform analytical findings from previous stages into an interactive and user-friendly visualization system.The dashboard enables dynamic exploration of global weather patterns, seasonal variations, regional comparisons, and extreme weather events using intuitive visual components and filters.

2. Tools & Technologies

- Python
- Streamlit
- Plotly Express
- Pandas
- Matplotlib & Seaborn

3. Dashboard Features & Interactivity

The dashboard includes the following interactive components:

Filters

- Country selection (single/multiple)
- Year range selection
- Seasonal filtering
- Metric selection (temperature, humidity, precipitation, wind speed)
- Top N country selector

Key Features

- Real-time filtering of data
- Dynamic chart updates
- Multi-tab navigation
- Insight annotations for each section

These features allow users to explore climate data flexibly and efficiently.

4. Visualization Components

5. 4.1 Dashboard Overview

The dashboard provides an intuitive interface with filters and multiple sections for exploring climate data.

Figure 1: Dashboard Overview

This view highlights key metrics such as average temperature, humidity, and rainfall.

4.2 Climate Trends Analysis

Line charts visualize long-term changes in temperature and other climate variables.

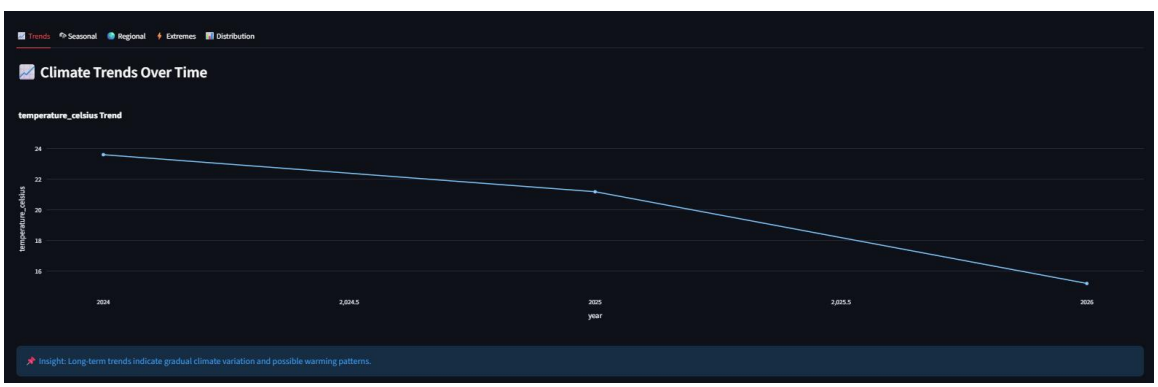

Figure 2: Climate Trends

Insight: Gradual variation and seasonal oscillations are observed.

4.3 Seasonal Analysis

Bar and line charts show how climate variables change across seasons.

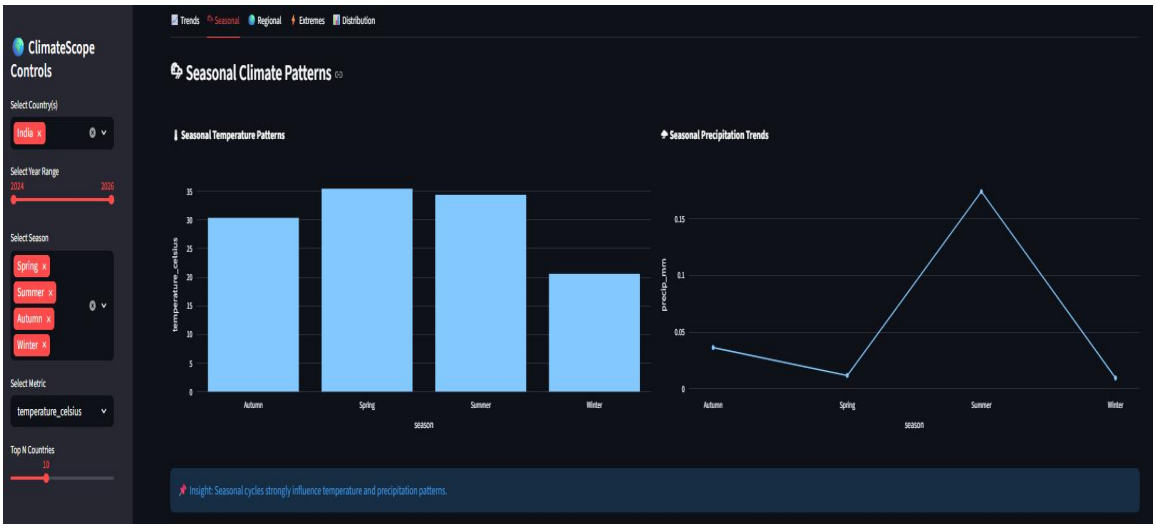

Figure 3: Seasonal Patterns

Insight: Summer has higher temperatures; precipitation varies regionally.

4.4 Regional Comparison

Geographical visualization compares climate across regions.

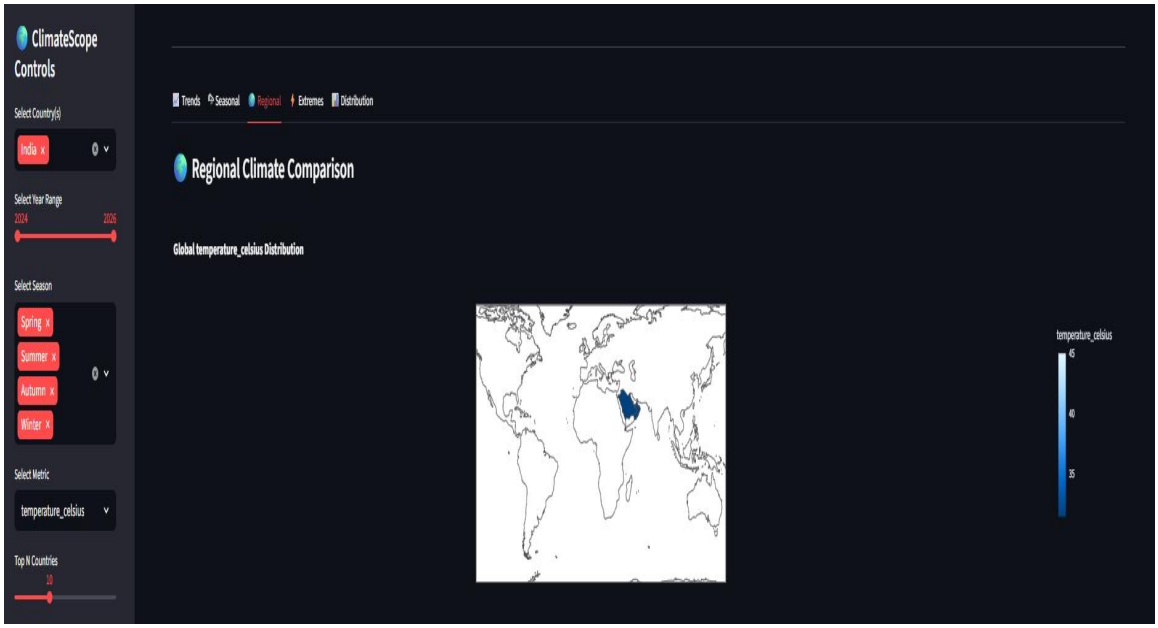

Figure 4: Regional Analysis

Insight: Climate differs significantly across locations.

4.5 Extreme Events

Scatter plots identify extreme weather events.

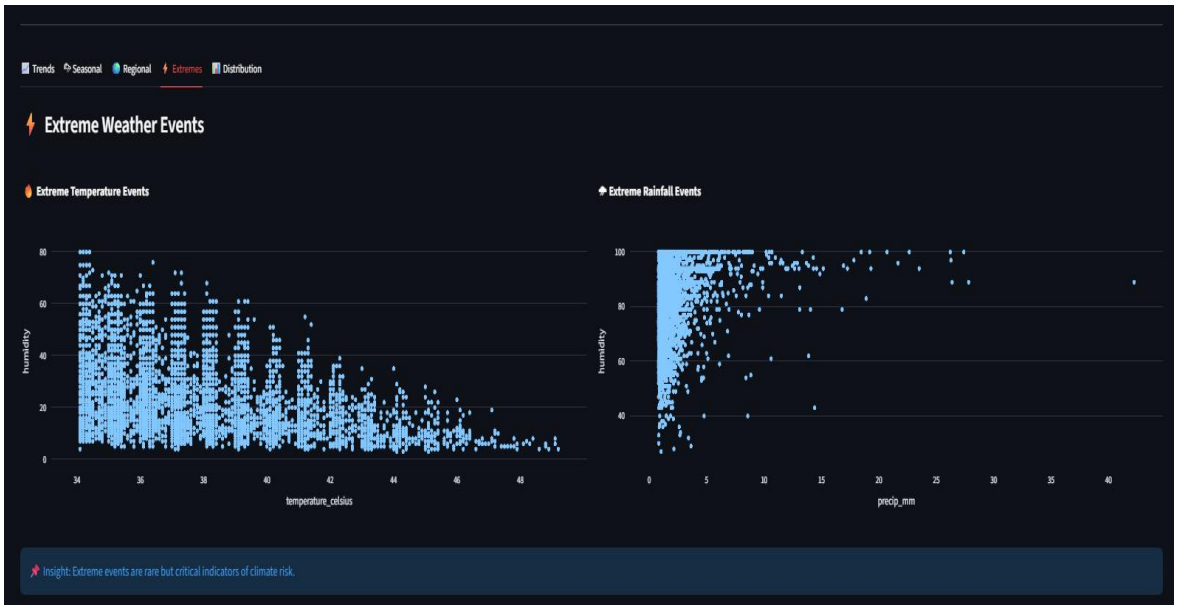

Figure 5: Extreme Events

Insight: Extreme values are rare but concentrated in specific areas.

4.6 Distribution Analysis

Histograms show the spread of temperature and precipitation.

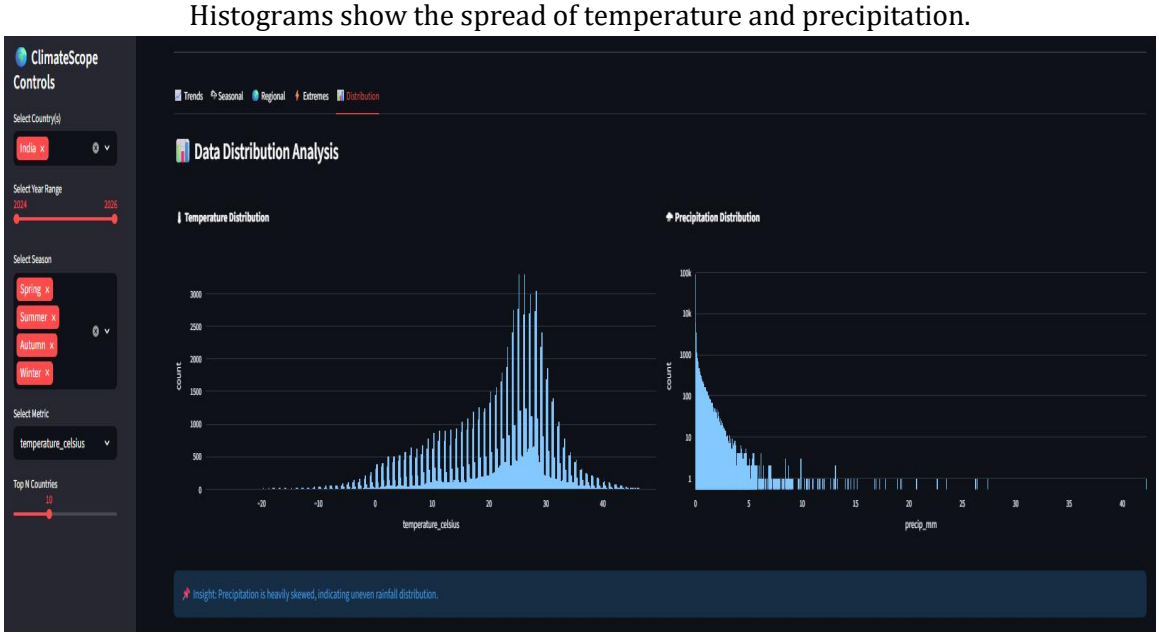

Figure 6: Distribution Analysis

Insight: Precipitation is right-skewed with frequent low values.

5. Correlation Analysis

A heatmap shows relationships between variables.

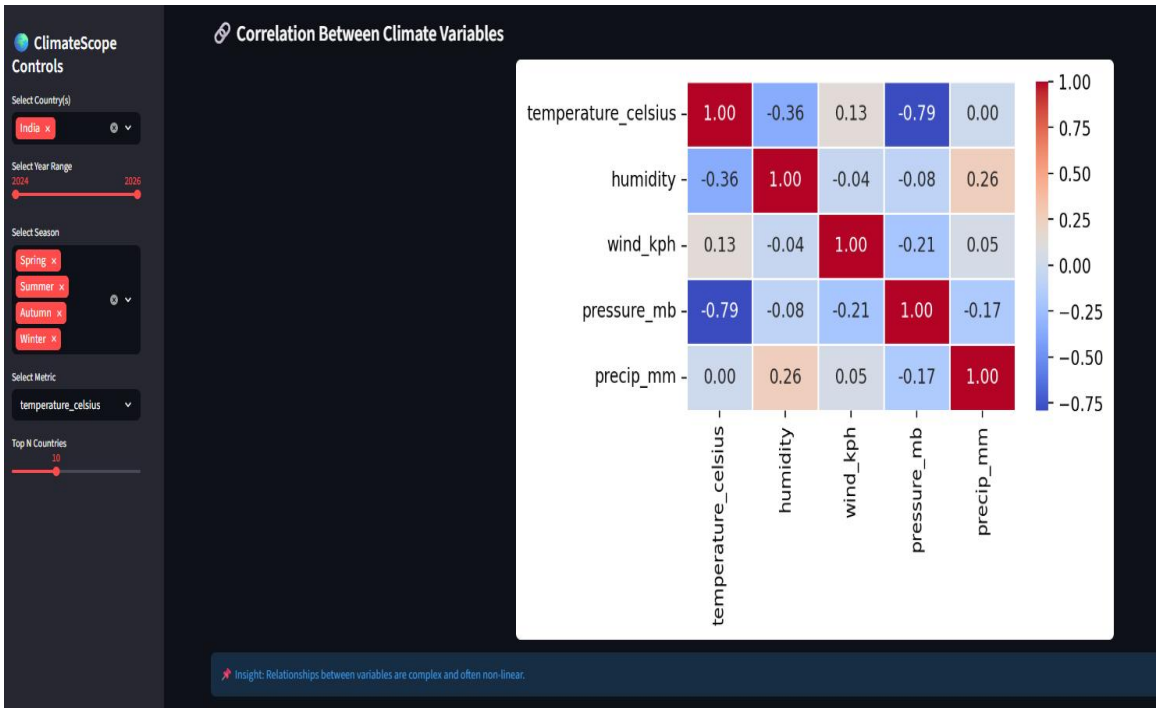

Figure 7: Correlation Heatmap

Insight: Weak to moderate relationships exist among variables.

6. Conclusion

Milestone 3 was successfully completed with the development of a fully functional and interactive dashboard. The dashboard effectively integrates multiple visualization techniques and interactive controls to provide meaningful insights into global climate patterns and extreme weather events.

This implementation serves as a strong foundation for final testing, reporting, and deployment in the next phase.

Next Steps

- Perform final testing and optimization
- Prepare comprehensive final report
- Deploy dashboard (optional)
- Add advanced features for future enhancement